Few online mistakes imply a small Boolean circuit for explicit circuit-induction rules

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Abstract

We ask when few online prediction mistakes imply that a Boolean target has a small circuit. Complexity is the number of two-input NAND gates, with no depth restriction. Inputs are presented one at a time, every label is revealed after prediction, and the conclusions require complete revelation of the domain whose complexity is measured.

We prove converses for three specified learning rules. A learner that always selects a smallest consistent circuit makes at least $C(f)/(3d+3)$ mistakes on the full d-dimensional Boolean cube. For Bayesian prediction by a fresh posterior sample each round, a geometric gate-count prior with uniform wiring gives $C(f) = O(dM \log(d + dM))$, where M is expected total mistakes. A prefix circuit language with explicit exception constructors improves this to $O(dM + 1)$ while retaining the usual $O(C(f) \log(d + C(f)) + 1)$ upper bound. Both Bayesian arguments use injective, inexpensive repairs of circuit descriptions to control log loss by sampling mistakes. The results allow any complete reveal order and infinite iid sampling on any fixed support, with complexity measured on that support. They rely on the specified learning rules and priors; the Bayesian proof does not give a converse for majority prediction. The main finite and infinite-stream statements have Lean proofs for the concrete circuit models and priors.

1 Setup and what a converse can say

Fix $d \geq 1$ and $\mathcal{X} = \{0,1\}^d$. A deterministic target is $f : \mathcal{X} \to \{0,1\}$. A circuit consists of a topologically ordered list of two-input NAND gates. Its available initial wires are the d input bits and constants $0, 1$. Each gate chooses an ordered pair of earlier wires, with repetitions allowed. The output is the last gate; the empty list outputs the constant zero. Dead gates are permitted. Its size $|c|$ is its number of gates. Let

$$C_A(f) = \min\{|c| : c(x) = f(x) \text{ for every } x \in A\}, \qquad C(f) = C_{\mathcal{X}}(f).$$

All these minima exist. Our conclusions concern total NAND gate count, with no fixed-depth restriction.

At round t, an input x_t is selected, the learner predicts, and the correct label $f(x_t)$ is revealed. Repetitions have the same label. The input may depend on past predictions and labels, but it must be selected before the fresh random prediction on that round. Write $A_t = \{x_1, \ldots, x_t\}$. A complete reveal of A means that precisely points from A have been queried and every point of A has appeared at least once. Inputs outside A need not be learned.

Coverage is unavoidable: labels on a small set of easy inputs cannot determine the target's complexity elsewhere. Likewise, an arbitrary learner may hard-code a highly complex target and

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make no mistakes. The theorems below concern specified circuit-induction procedures and their complete observations, not a representation theorem for every successful predictor.

We study three ways to select predictions from circuits consistent with the revealed labels. The first chooses a circuit of minimum gate count. The other two use exact Bayesian conditioning and draw a fresh circuit from the resulting posterior at each round. They differ only in their priors on circuit descriptions. The geometric prior penalizes both gate count and the addresses of each gate's inputs; the prefix prior gives a fixed encoding cost to an exception at one specified input.

A common repair operation drives all three arguments. First we construct that operation and prove the deterministic converse. We then express Bayesian learning as cumulative log loss and bound that loss using the repair cost under each prior. The final sections extend the conclusions to iid inputs and explain the roles of coverage, prediction rule, and circuit representation.

2 A one-point circuit repair

The repair retains the original circuit and appends gates that change its answer at one chosen input. It leaves every other answer intact. Keeping the original gate list also lets us recover the source description, which is essential when summing Bayesian prior masses. Put $R_d = 3d + 3$.

Lemma 1 (Injective overwrite)**.** *For each $(x, y) \in \mathcal{X} \times \{0, 1\}$ there is a map $P_{x,y} : \mathcal{C} \to \mathcal{C}$ that appends exactly $r_x \le R_d$ gates and obeys*

$$P_{x,y}(c)(z) = \begin{cases} y, & z = x, \\ c(z), & z \neq x. \end{cases}$$

The number r_x is independent of the source circuit, and the map is injective on circuit descriptions.

Proof. Starting from the constant-one wire, form the equality indicator $e_x(z) = \bigwedge_{j=1}^{d} \mathbf{1}\{z_j = x_j\}$. A negative literal requires one NAND gate; conjoining the next literal to the accumulator requires two. This costs $2d + \#\{j : x_j = 0\} \le 3d$ gates. If $y = 0$, compute $c(z) \wedge \neg e_x(z)$ in three further NAND gates. If $y = 1$, compute $c(z) \vee e_x(z)$ in three further gates. The original output wire is available in both cases, including the empty circuit's constant-zero output. Deleting the final $r_x = 2d + \#\{j : x_j = 0\} + 3$ gates recovers the original ordered list, proving injectivity. No simplification of dead or redundant gates is performed. □

For a noiseless history, overwriting the current input with its true label preserves all earlier labels. This remains true for repeated inputs. Injectivity is a statement about descriptions, not functions: two different functions can coincide after their value at one point is overwritten. Distinguishing these facts matters for prior-mass estimates.

3 A linear converse for minimum-consistent-circuit prediction

Theorem 2 (Deterministic Occam converse)**.** *At every round let the learner predict with any minimum-size circuit consistent with the preceding observations, with arbitrary tie-breaking. If it has made B_t mistakes through round t, then*

$$C_{A_t}(f) \le R_d B_t.$$

In particular a complete reveal of A entails at least $C_A(f)/R_d$ mistakes, and a complete reveal of the cube entails at least $C(f)/R_d$ mistakes.

Proof. Initially the minimum consistent size is zero. Adding a constraint cannot decrease that minimum. If the current prediction is correct, the selected minimum-size circuit remains consistent, so the minimum does not change. If the prediction is wrong, Lemma 1 produces a consistent circuit with at most R_d additional gates. Induction proves the result. □

This is a converse, not a polynomial upper bound on the mistakes of arbitrary minimum-size tie-breaking. The proof does not assert that a wrong prediction eliminates a substantial fraction of competing circuits. Exact minimum-circuit selection can be computationally intractable; no efficient implementation is claimed.

4 Bayesian sampling, log loss, and complete reveals

Let π be a prior on a countable set of circuit descriptions, with every finite noiseless history assigned positive mass. Let

$$E_t = \{c : c(x_i) = f(x_i)\ (1 \le i \le t)\}, \qquad Z_t = \pi(E_t), \qquad L_t = -\log Z_t.$$

The posterior is $\pi(\cdot \mid E_{t-1})$. Define

$$p_t = \pi\{c(x_t) = f(x_t) \mid E_{t-1}\}, \quad q_t = 1 - p_t, \quad m_t = \sum_{i=1}^{t} q_i.$$

A posterior-sampling predictor draws a fresh circuit from that posterior and outputs its label. Its conditional mistake probability is exactly q_t. For a fixed input sequence, m_t is its expected number of mistakes. For an adaptive or random input sequence, m_t is the random sum of these conditional probabilities, and expected actual mistakes are $\mathbb{E}m_t$ (or the analogous stopped/infinite sum).

Irrespective of the query order, exact conditioning gives

$$L_t = \sum_{i=1}^{t} -\log p_i, \qquad m_t \le L_t. \tag{1}$$

On a complete reveal of A, the final value is the deterministic quantity

$$L_A = -\log \pi\{c : c|_A = f|_A\}.$$

The forward inequality $m_t \le L_t$ does not by itself give a converse: one incorrect prediction can correspond to an arbitrarily unlikely correct label and hence arbitrarily large log loss. The next sections prevent this by repairing descriptions at controlled prior cost.

5 A natural geometric gate-count prior

Define π_d by first drawing S with $\mathbb{P}(S = s) = 2^{-(s+1)}$, $s = 0, 1, \ldots$. Conditional on $S = s$, draw the ordered inputs to gate i independently and uniformly from its $d + i + 1$ earlier wires. Thus each particular circuit c of size s has

$$\pi_d(c) = 2^{-(s+1)} \prod_{i=1}^{s} (d + i + 1)^{-2}. \tag{2}$$

This is a normalized probability distribution: the product counts the inverse of the number of wiring lists of length s. It is a prior on descriptions and induces a prior on functions by summing their masses.

For $u \geq 0$, put

$$A_d(u) = \log(4/3) + R_d \log 2 + 2R_d \log(u/\log 2 + 4d + 6). \tag{3}$$

Lemma 3 (A lower bound on predictive label mass)**.** *Under* (2)*, any noiseless history and any next input satisfy*

$$p_t \geq \exp[-A_d(L_{t-1})].$$

Consequently every finite history satisfies

$$L_t \leq 2A_d(L_t)m_t. \tag{4}$$

Proof. Set $K = \lceil L_{t-1}/\log 2 \rceil + 1$. Since $\pi_d(S > K) = 2^{-(K+1)}$, the posterior mass on $S > K$ is at most $2^{-(K+1)}e^{L_{t-1}} \leq 1/4$. At least $3/4$ of posterior mass therefore lies on consistent source circuits of size at most K.

Repair every such source using $P_{x_t, f(x_t)}$. The resulting descriptions are distinct, fit all the observations including the new one, and have prior-to-source probability ratio

$$2^{-r_{x_t}} \prod_{i=s+1}^{s+r_{x_t}} (d+i+1)^{-2} \geq \left[2(d+K+R_d+1)^2\right]^{-R_d}.$$

Summing over the injective image and dividing by the preceding normalizer gives the lower bound on p_t, because $d + K + R_d + 1 \leq L_{t-1}/\log 2 + 4d + 6$.

For $0 < b \leq p \leq 1$,

$$-\log p \leq \frac{-\log b}{1-b}(1-p).$$

Indeed, $(-\log p)/(1-p)$ decreases with p, with the continuous value one at $p = 1$. The function A_d is increasing and exceeds $\log 2$. Use the common lower bound $b = e^{-A_d(L_t)} \leq 1/2$ for all rounds up to t and sum. This gives (4). □

Theorem 4 (Geometric circuit-prior converse)**.** *Suppose posterior sampling with prior* (2) *completely reveals $A \subseteq \mathcal{X}$. Let M be its expected total number of mistakes, including any randomness in a predictable query order. Put $C = C_A(f)$. Then*

$$M \geq \frac{C \log 2}{2A_d(C \log 2)}, \tag{5}$$

$$C \leq \frac{12R_dM}{\log 2} \log\left(4d + 6 + \frac{12R_dM}{\log 2}\right). \tag{6}$$

In particular $C = O(dM \log(d + dM))$. Conversely,

$$M \leq (C+1)\log 2 + 2C \log(d + C + 1). \tag{7}$$

All constants are independent of the reveal order and of $|A|$.

Proof. Any circuit agreeing with f on A has at least C gates, so $\pi_d(E_A) \leq \pi_d(S \geq C) = 2^{-C}$ and $L_A \geq C \log 2$. The function $u/A_d(u)$ is increasing on $[0, \infty)$: its derivative has numerator

$$A_d(u) - uA_d'(u) = \log(4/3) + R_d \log 2 + 2R_d \left[\log v - \frac{v - (4d+6)}{v}\right] > 0, \quad v = u/\log 2 + 4d + 6 \geq 10.$$

Equation (4) on each complete-reveal path therefore gives

$$m \geq \frac{L_A}{2A_d(L_A)} \geq \frac{C \log 2}{2A_d(C \log 2)}.$$

Taking expectations proves (5).

Since $C + 4d + 6 \geq 10$ and $R_d \geq 6$, $A_d(C \log 2) \leq 3R_d \log(C + 4d + 6)$. Hence $C \leq a \log(b + C)$ with $a = 6R_d M / \log 2$ and $b = 4d + 6$. For $a > 0, b \geq e$, the implication

$$x \leq a \log(b + x) \quad \Longrightarrow \quad x \leq 2a \log(b + 2a)$$

follows by setting $x_0 = 2a \log(b + 2a)$, observing $b + x_0 \leq (b + 2a)^2$, and using the increase of $x - a \log(b + x)$ for $x \geq x_0$. This proves (6); if $M = 0$, (5) already forces $C = 0$.

For the upper bound, choose a minimum consistent circuit of size C. Its probability in (2) is at least $2^{-(C+1)}(d + C + 1)^{-2C}$. Thus $m \leq L_A \leq (C + 1) \log 2 + 2C \log(d + C + 1)$ on every complete path. Taking expectations gives (7). □

Remark 5 (Partial histories)**.** *Before full revelation, (4) still holds, and applying the same reasoning gives a small circuit consistent with the observed points in terms of m_t. It bounds $C_{A_t}(f)$, not the unrestricted $C(f)$. No inference about unobserved labels is implicit.*

6 A prefix circuit prior with a sharper converse

The logarithmic factor in Theorem 4 comes from the wire addresses of appended gates: an exception appended to a longer circuit costs more prior probability. An explicit exception constructor removes this dependence. The following is a precise prior, not an assertion that all circuit encodings have this property.

Use the following prefix language, with d known to the decoder.

- The string `00` denotes constant zero.
- A string `01`b denotes a NAND circuit with $s \geq 1$ gates. The prefix b starts with the unary length 1^s0, followed, for each gate i, by two wire addresses of length $\lceil \log_2(d + i + 1) \rceil$. Out-of-range addresses are invalid.
- A string `1`xyp denotes the overwrite of program p at input x with label y, where x is encoded in exactly d bits, y in one bit, and p is another valid program.

The valid finite strings are prefix-free: the initial tags separate the cases, unary lengths determine base-code lengths, and the recursive case has fixed-size metadata before its single prefix-free child. Every program denotes a Boolean function and every Boolean circuit has such a description. A program of bit length ℓ compiles into a NAND circuit of at most 3ℓ gates, by Lemma 1 and $R_d \leq 3(d + 2)$.

For valid strings p, define

$$\widetilde{\pi}_d(p) = Z_d^{-1} 2^{-2\ell(p)}, \qquad Z_d = \sum_{p \text{ valid}} 2^{-2\ell(p)}. \tag{8}$$

Kraft's inequality and the code `00` imply $1/16 \leq Z_d \leq 1$. Thus this is a proper circuit-description prior. Write $K_A(f)$ for the shortest program length agreeing with f on A.

Theorem 6 (Linear converse with exception codes)**.** *For posterior sampling with* (8) *and a complete reveal of A, expected mistakes M satisfy*

$$C_A(f) \le \frac{6(d+2)}{1-2^{-2(d+2)}} M + 12.$$

Moreover $M \le 2K_A(f)\log 2$, and $K_A(f) = O(C_A(f)\log(d + C_A(f)) + 1)$. Consequently this same prior has both an $O(dM + 1)$ converse and the usual circuit-size upper bound.

Proof. The map $p \mapsto \mathtt{1}xyp$ is injective, preserves earlier consistent labels when $y = f(x)$, and adds exactly $d + 2$ bits. Its prior ratio is the constant $\beta = 2^{-2(d+2)}$. Applying it to all descriptions in the preceding version space gives predictive correct-label probability at least β. Therefore

$$L_A \le \frac{2(d+2)\log 2}{1-\beta}\, m$$

on every complete path.

For any real $r \ge 0$, Kraft's inequality gives

$$\widetilde{\pi}_d\{\ell(p) \ge r\} \le Z_d^{-1} 2^{-r} \sum_p 2^{-\ell(p)} \le 16\, 2^{-r}.$$

Every program agreeing with f on A has length at least $C_A(f)/3$, since its compiled circuit has at most three times its length in gates. It follows that $L_A \ge (C_A(f)/3)\log 2 - \log 16$. Combine the two inequalities and take expectations for the asserted converse. A shortest fitting program has probability $2^{-2K_A(f)}/Z_d$, so $m \le L_A \le 2K_A(f)\log 2 + \log Z_d \le 2K_A(f)\log 2$. Finally a circuit of size $s \ge 1$ has a base description of length

$$s + 3 + 2\sum_{i=1}^{s} \lceil \log_2(d+i+1) \rceil \le s + 3 + 2s\lceil \log_2(d+s+1) \rceil.$$

For $s = 0$, use $\mathtt{00}$. This proves the forward complexity estimate. □

The squared code weight in (8) is useful: prefix freeness alone does not give an exponentially decreasing tail for the ordinary weights $2^{-\ell(p)}$. The additional length penalty makes the tail estimate uniform over this language. The result is about the specified prior, not an invariant property of all "universal" priors.

7 Random input orders and iid sampling

All results above hold for any fixed reveal order. The Bayesian lower bounds were established pathwise for the sum of conditional mistake probabilities, using a final log loss depending only on the revealed set. Consequently the same bounds hold after averaging over random orders, including orders selected using previous sampled predictions. The input for the current round must still precede that round's sample.

Corollary 7 (Infinite iid input sequence)**.** *Let D be any distribution on $\mathcal{X}$ and $A = \{x : D(x) > 0\}$. Run any of the specified learners on an infinite iid input sequence from D. Every point of A is eventually seen almost surely. Theorem 2 holds for its total actual mistakes, and Theorems 4 and 6 hold for expected total posterior-sampling mistakes, with target complexity $C_A(f)$. When D has full support this is $C(f)$.*

Proof. The set A is finite, and each of its points has a finite first-occurrence time almost surely. After all have appeared, every consistent predictor is correct on all future draws. Predictions at repetitions are already correct before this stopping time as well. Thus there are at most $|A|$ possible mistake rounds. Conditional expectation and Tonelli's theorem identify expected actual mistakes with the expected sum of conditional mistake probabilities; the complete-reveal results then apply. □

No lower bound on the positive probabilities $D(x)$ is required for total mistakes. Such a bound would matter for how long it takes to reveal the support: very rare inputs can postpone the decisive mistakes for arbitrarily many rounds. A small asymptotic error rate or a small number of mistakes over a finite horizon is therefore a different claim from a small total number of mistakes over the infinite sequence.

8 Limits of the converse

Arbitrary learners. For any prescribed function f, a learner that always returns $f(x)$ makes zero mistakes. By counting NAND circuits, there are Boolean functions with circuit size of order $2^d/d$. Thus no bound in terms of mistakes and d alone holds for arbitrary learners without restricting their initial knowledge or representation. The same observation applies to an arbitrarily selected prior: a point mass on a complex f makes zero posterior-sampling mistakes. Even full support does not repair this. For example, assigning mass $1-\eta$ to f and spreading the remaining η over all Boolean functions gives cumulative sampling mistakes at most $-\log(1-\eta)$, however complex f is.

Unrevealed inputs. For $d \geq 2$, a learner can observe only a point where $f = 0$, while the remaining truth table has almost arbitrary circuit complexity. Taking the input distribution to be concentrated on that point gives no control of global $C(f)$. The valid complexity in the iid theorem is the minimum size needed to match f on the distribution's support.

Posterior majority. Sampling mistake probability is $1-p_t$, whereas majority prediction is correct whenever $p_t > 1/2$. A sequence with $p_t = 3/4$ on every round has no majority mistakes but positive sampling mistakes and log loss on every round. This can occur for a prior with independently biased truth-table bits. The mass-repair proofs therefore do not establish a majority converse, even though they establish one for posterior sampling. The separate deterministic Occam theorem concerns minimum-size consistent circuits, not posterior majorities.

A minimum description versus a function prior. The geometric and prefix priors here assign mass to descriptions, and multiple descriptions of the same function are all retained. Injectivity of repairs makes their masses add correctly. For a prior specified directly on functions, the overwrite map has up to two preimages and its weights must be checked separately. Merely declaring that the prior "prefers smaller circuits" does not imply the quantitative ratios or tail bounds used above.

Depth, noise, and computation. The conclusions count all NAND gates and allow their depth to grow. One-point overwrites do not preserve a fixed depth architecture. Repeated conflicting labels are excluded; without realizability the consistent posterior can be empty. These are statistical and representation statements, with no efficient circuit minimization or posterior sampler asserted.

Scope of the formal proofs. The Lean development proves the deterministic, geometric-prior, and prefix-prior converse theorems with their stated constants, including their infinite-iid versions. It uses actual NAND gate lists and the specified probability laws. Prefix programs are represented by abstract syntax with the stated lengths and proved Kraft counting; an executable bitstring serializer and parser are not included. The limitation examples in this section and the circuit-counting asymptotic are explanatory mathematical arguments outside that formalization.